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# Variable speed limit control method of freeway merging area in a car-truck mixed heterogeneous traffic environment

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ABSTRACT

To alleviate traffic congestion and enhance safety in freeway merging areas under car-truck mixed heterogeneous traffic environment, this study proposes a variable speed limit (VSL) control method. First, a lane-level cell transmission model (CTM) is developed to predict traffic states, accounting for four distinct vehicle types: human-driven cars (HCs), human-driven trucks (HTs), connected autonomous cars (CACs), and connected autonomous trucks (CATs). Second, a collision risk prediction model for the merging area is constructed uses traffic simulation data through count model fitting. An optimal VSL control function is formulated to enhance traffic efficiency and safety. This function and CTM are then integrated into a model predictive control (MPC) framework, establishing a comprehensive VSL control method. Finally, simulations conducted using Python and SUMO verify the effectiveness of the proposed method. The results show that, compared with the no-control scenario, the VSL method significantly reduces total travel time (TTT) and collision risk (TCR) at the macroscopic level. At the microscopic level, it increases the average speed and reduces travel delay, time-exposed time-to-collision (TET), and time-integrated time-to-collision (TIT), thereby further validating the effectiveness of the method. The proposed research framework offers theoretical insights and technical guidance for future freeway traffic management in a car-truck mixed heterogeneous traffic environment.

## 1. Introduction

Rapid progress in vehicle communication, autonomous driving decision-making and planning, and control execution has ushered connected autonomous vehicles (CAVs) into a new era of large-scale implementation. However, the complete replacement of human-driven vehicles (HVs) by CAVs remains unlikely in the near future [1]. Meanwhile, given the high-risk nature of freeway freight transport and the resulting decline in the number of truck drivers, connected autonomous trucks (CATs) are expected to have broader application prospects. As a consequence, heterogeneous traffic flows composed of human-driven cars (HCs), human-driven trucks (HTs), connected autonomous cars (CACs), and CATs are projected to coexist on freeways in the foreseeable future. Due to disparities in perception–decision–control capabilities between CAVs and HVs, along with variations in dynamic performance between passenger cars and trucks, the complexity of the traffic environment is substantially heightened. This complexity is particularly pronounced in freeway merging areas, where frequent speed variations, lane changes, and vehicle interweaving contribute to decreased traffic efficiency and elevated collision risk.

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As a key active traffic management (ATM) strategy on freeways, the variable speed limit (VSL) method has seen growing implementation in recent years [2]. By capping maximum speeds, VSL establishes a high-density, low-speed control zone that delays vehicle entry into bottlenecks and reduces speed differentials, thereby enhancing freeway efficiency and safety. However, variability in driver compliance with VSL directives can undermine the expected effectiveness of the control strategy. In car-truck mixed heterogeneous traffic environment, CAVs exhibit superior controllability and compliance, thereby mitigating the disruptive influence of HVs. As a result, VSL proves more feasible and effective in heterogeneous traffic environments than in conventional HV-only scenarios. Accordingly, investigating VSL control methods for freeway merging areas in a car-truck mixed heterogeneous traffic environment is crucial, as it offers substantial practical value for improving traffic management and control in these critical areas.

Recent studies have investigated this topic extensively. For example, Jin et al. [3] and Zhang et al. [4] utilized macroscopic traffic flow models, such as METANET and the cell transmission model (CTM), to predict the temporal evolution of heterogeneous traffic conditions. Based on these models, they developed VSL control strategies grounded in optimal control and model predictive control (MPC), incorporating multi-objective optimization functions that simultaneously consider traffic efficiency, safety, and environmental sustainability. In addition, Vrbanić et al. [5–6] and Wu et al. [7] explored VSL strategies under different CAV penetration rates using reinforcement learning (RL) algorithms, including Q-learning, deep Q-networks (DQN), and deep deterministic policy gradients (DDPG). However, most existing studies address heterogeneous traffic dominated by passenger cars, often neglecting trucks—a critical vehicle type on freeways. The presence of trucks increases speed differentials and intensifies traffic flow heterogeneity, thereby elevating the risks of congestion and collisions. Accurately estimating collision risk in heterogeneous traffic scenarios remains a significant challenge. Therefore, developing a VSL control method for freeway merging areas that effectively balances traffic efficiency and safety in a car–truck mixed heterogeneous traffic environment is of significant importance.

This study addresses these research gaps, and its main contributions are as follows:

a) A lane-level CTM is developed to simulate the evolution of traffic dynamics in freeway merging areas under a car–truck mixed heterogeneous traffic environment, enabling short-term traffic state prediction.
b) Based on simulation data, a collision risk prediction model for merging areas is constructed using a count model fitting approach.
c) A dual-objective optimization function that jointly considers traffic efficiency and safety is formulated and integrated with the CTM into an MPC framework, resulting in a VSL control method for freeway merging areas.

The remainder of this paper is organized as follows. Section 2 reviews the literature review. Section 3 describes the research scenario and methodology. Section 4 details the proposed VSL control method. Section 5 discusses the simulation experiments and results. Section 6 concludes the paper.

## 2. Literature review

Research on VSL control primarily focuses on accurate real-time prediction of traffic flow dynamics to ensure timely control and the selection of appropriate VSL control methods to enhance traffic management effectiveness. Existing VSL methods mainly fall into feedback control and RL-based algorithms. To investigate VSL control methods in traffic environments involving CAVs, the literature is reviewed from three perspectives: (1) short-term traffic state prediction and modeling, (2) VSL strategies based on feedback control, and (3) VSL strategies based on RL.

### *2.1. Short-term traffic state prediction and modeling*

Accurate short-term traffic state prediction is fundamental to effective VSL implementation. Qian et al. [8] developed a macroscopic heterogeneous traffic flow model based on the CTM, incorporating interactions among different vehicle types, and validated it using numerical simulations and NGSIM data. Qin et al. [9], Wu et al. [10], and Jin et al. [11] derived a CTM suitable for heterogeneous traffic flow based on microscopic car-following (CF) models of HVs and CAVs and validated its effectiveness through simulation. Laan et al. [12] and Zhang et al. [13] developed a multi-lane CTM incorporating CAV-dedicated lanes (CAVLs) to analyze the impact of the number, width, and placement for CAVLs on heterogeneous traffic flow. In the domain of trajectory-level forecasting, Lin et al. [14] used long short-term memory (LSTM) models to predict HV trajectories in a heterogeneous traffic environment, and Wang et al. [15] combined Double DQN with spatiotemporal information extraction to anticipate traffic evolution and develop CAV-based merging strategies.

### *2.2. The VSL strategies based on feedback control*

VSL strategies based on feedback control primarily monitor current traffic conditions and forecast future states. Speed limits are dynamically adjusted in real-time to optimize traffic state, enabling rapid and adaptive responses to evolving conditions. Mao et al. [16] enhanced the CTM by introducing variable cell lengths to more accurately represent the dynamic characteristics of VSL areas in connected vehicle environments. They implemented VSL control within an MPC framework. Yu et al. [17] developed a CTM to simulate HVs, heavy-duty vehicles, and autonomous vehicles (AVs). They proposed a VSL strategy via multi-objective optimization to improve overall traffic efficiency and enable smoother speed transitions. Similarly, Chen et al. [18] extended the application of CTM and proposed a VSL strategy aimed at reducing energy consumption. Wang et al. [19] employed the METANET model to forecast traffic states and proposed a VSL optimization strategy tailored for freeway mainlines.

In addition, Khondaker et al. [20] formulated a multi-objective optimization function that balances traffic efficiency, safety, and environmental sustainability by incorporating microscopic driver behavior models. They employed MPC to address the VSL control problem in the AV environment. Han et al. [21] introduced a VSL control based on the discrete Lagrangian model, using the average speed of a CAV platoon as the control variable to mitigate freeway congestion waves. Other studies, including those by Grumert et al. [22], Han et al. [23], and Li et al. [24–25], developed VSL control systems with varying levels of complexity and functionality, and validated their effectiveness in improving traffic flow stability, safety, and efficiency. Jin et al. [26] proposed a lane-specific dynamic VSL control strategy, wherein speed limits are configured for individual lanes to guide vehicles in making timely lane changes before reaching bottlenecks, alleviating congestion, and reducing capacity loss.

### 2.3. The VSL strategies based on reinforcement learning

With advanced autonomous driving technologies, RL has gained prominence in VSL research. Han et al. [27] and Li et al. [28] developed RL-based VSL control methods to mitigate freeway congestion waves and reduce collision risks. Gregurić et al. [29] introduced a spatial VSL area concept and employed the DDPG algorithm to manage back-propagating congestion. Ke et al. [30] proposed a VSL control method integrating transfer learning with the DQN algorithm to minimize total travel time in freeway bottleneck areas. Wu et al. [6] introduced a lane-level differential VSL control strategy to enhance traffic efficiency and safety while reducing emissions using the deep actor-critic algorithm.

Other researchers (e.g., Vrbanić et al. [4,5] and Kušić et al. [31]) utilized CAVs as mobile sensors and actuators, combining speed transition matrices with RL to form adaptive VSL systems for heterogeneous traffic. Cheng et al. [32] designed a cooperative VSL model via DDPG to improve mainline–ramp merging coordination. Pan et al. [33] developed a joint framework integrating VSL, ramp metering, and lane-changing using model-based RL. Lu et al. [34] advanced lane-level control strategies that couple VSL with lane selection, achieving notable travel time improvements.

In summary, significant progress has been made in VSL control under traffic conditions involving CAVs; however, several challenges remain. First, existing research predominantly focuses on heterogeneous traffic flow dominated by passenger cars, with insufficient consideration given to the role of heavy-duty vehicles such as trucks. The evolution of traffic operations under mixed car-truck conditions is also inadequately addressed, complicating the development of reasonable and accurate VSL control strategies for such environments. Second, large speed differentials in merging areas and frequent lane-change behavior increase the risk of collisions. However, research on collision risk prediction under mixed car-truck traffic conditions remains limited and requires further development. Third, most existing models pay limited attention to the balanced optimization of both traffic efficiency and safety. Therefore, in managing freeway merging areas under car-truck mixed heterogeneous traffic conditions, designing an efficient and flexible VSL control strategy that adapts to dynamic traffic conditions and simultaneously enhances traffic operation efficiency and safety remains a critical research challenge. Given the limited generalization ability, poor interpretability, and high training costs associated with reinforcement learning, this study adopts a VSL control method based on model predictive control (MPC).

## 3. Research scenarios and method framework

This study focuses on the merging areas of a one-way, two-lane freeway, as shown in Fig. 1. The vehicle types are classified as HCs, HTs, CACs, and CATs based on the composition of freeway traffic. Specifically, CACs and CATs communicate with the roadside unit (RSU). They are centrally coordinated through it, whereas HCs and HTs are monitored by the RSU using sensors such as cameras and millimeter-wave radars. In car-truck mixed heterogeneous traffic environment, various vehicles merge and interact from the mainline and the onramp within the merging area. Frequent speed fluctuations and lane-changing behaviors result in substantial speed differentials within the traffic flow, thereby inducing congestion bottlenecks and increasing collision risks.

To enhance traffic efficiency and safety within the merging area, this study employs the freeway RSU to collect real-time traffic

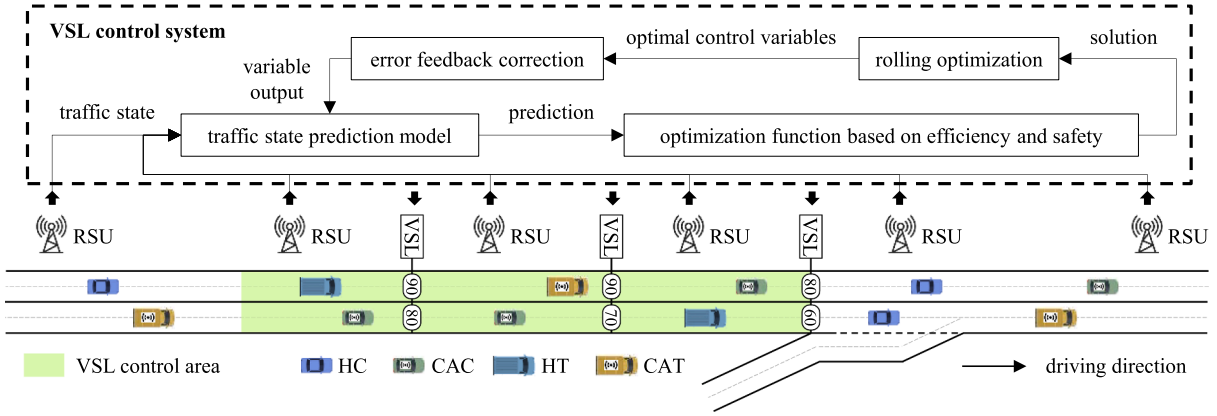



**Fig. 1.** Research scenario and method framework.

operation data from the merging area and develops a CTM tailored to car-truck mixed heterogeneous traffic flow for short-term traffic prediction. An objective optimization model is formulated to determine the optimal speed limit to optimize traffic efficiency and safety. The CTM and VSL optimization models are integrated into an MPC framework. Based on this framework, speed limits are dynamically adjusted using rolling horizon optimization to achieve the desired operational objectives. The calculated speed limits are communicated to the traffic system via variable message signs and vehicle-to-infrastructure (V2I) communication, guiding both HVs and CAVs to comply, thereby realizing closed-loop feedback control.

## 4. Variable speed limit control method

### 4.1. Freeway merging area CTM for car-truck mixed heterogeneous traffic flow

#### 4.1.1. The fundamental diagram of traffic flow

In car-truck mixed heterogeneous traffic flow, the CAV penetration rate is denoted as $p_A$ ($0 \le p_A \le 1$), and the truck mixing rate is denoted as $p_T$ ($0 \le p_T \le 1$). Accordingly, the proportion of HCs is $p_{HC} = (1\text{-}p_A) \times (1\text{-}p_T)$, and the proportion of HTs is $p_{HT} = (1\text{-}p_A) \times p_T$. The proportion of CACs is $p_{CAC} = p_A \times (1\text{-}p_T)$, and the proportion of CATs is $p_{CAT} = p_A \times p_T$.

Given that CAVs possess vehicle-to-vehicle (V2V) communication capabilities and autonomous driving functionality, a CAV following another CAV can adopt cooperative adaptive cruise control (CACC) mode for platoon driving. When a CAV follows an HV, it can only adopt adaptive cruise control (ACC) mode due to the absence of communication and coordination capabilities [35]. Moreover, CACs and CATs are capable of cooperative platoon driving. According to [36], the maximum formation size ($R$) is defined for CAV platooning. When the downstream CAV platoon reaches this limit, a CAV automatically becomes the leader of a new platoon. Based on the stochastic spatial distribution of different vehicle types, the classifications and corresponding distance headway for HCs, HTs, CACs, and CATs are specified, as shown in Fig. 2.

As depicted in Fig. 2, 20 distinct CF types exist within a car-truck mixed heterogeneous traffic environment. Specifically, when the target vehicle is an HC, it may follow HC, HT, CAC, or CAT, resulting in 4 CF types, with the corresponding headways denoted as $H_{HC\text{–}HC}$, $H_{HT\text{–}HC}$, $H_{CAC\text{–}HC}$, and $H_{CAT\text{–}HC}$. Similarly, when the target vehicle is an HT, the associated headways are represented as $H_{HC\text{–}HT}$, $H_{HT\text{–}HT}$, $H_{CAC\text{–}HT}$, and $H_{CAT\text{–}HT}$. When the target vehicle is a CAC, considering that CAC and CAT can cooperate in platoon formation, coupled with the constraint on maximum platoon size, the CAC may follow HC, HT, the trailing vehicle of a CAV platoon (either CAC or

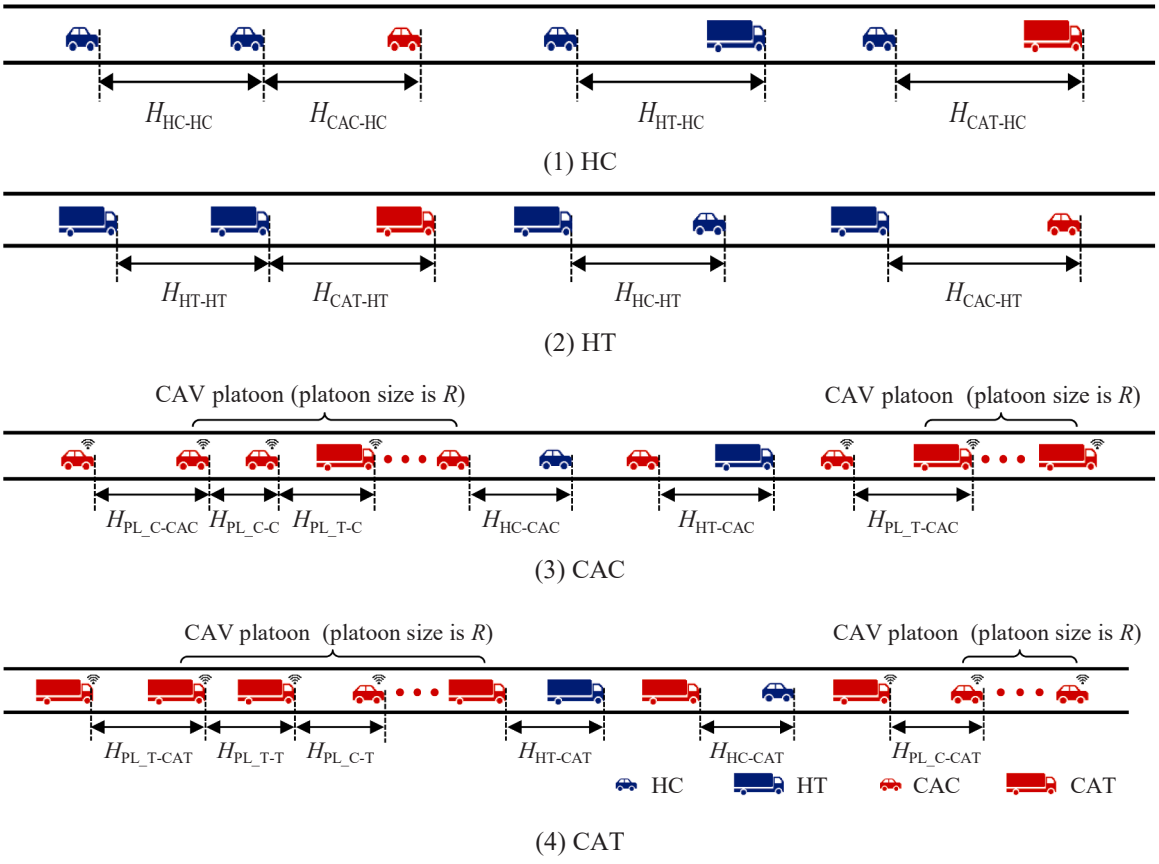



**Fig. 2.** CF type and its headway.

CAT), or a non-trailing vehicle within the CAV platoon (either CAC or CAT). This results in 6 CF types, with corresponding headways denoted as $H_{\text{HC–CAC}}$, $H_{\text{HT–CAC}}$, $H_{\text{PL_C–CAC}}$, $H_{\text{PL_T–CAC}}$, $H_{\text{PL_C–C}}$, and $H_{\text{PL_T–C}}$. similarly, when the target vehicle is a CAT, the associated headways are represented as $H_{\text{HC-CAT}}$, $H_{\text{HT-CAT}}$, $H_{\text{PL_C-CAT}}$, $H_{\text{PL_T-CAT}}$, $H_{\text{PL_C-T}}$, and $H_{\text{PL_T-T}}$.

In car–truck mixed heterogeneous traffic flows, vehicles exhibit spatial randomness, while CAVs tend to cluster due to their platooning capabilities. A Markov chain model is employed to characterize the spatial distribution of vehicles, with an aggregation factor $O$ $(-1 \le O \le 1)$ introduced to describe the platooning tendency of CAVs. Based on this model, the probabilities of various CF types are quantified. The state space of the Markov chain is defined as {CAC, CAT, HC, HT}, with the initial probability distribution given by $\{p_{\text{CAC}}, p_{\text{CAT}}, p_{\text{HC}}, p_{\text{HT}}\}$. CAC and CAT are capable of coordinating in platoon formation. Thus, CAV platooning behavior is independent of whether the vehicle is a passenger car or a truck. Let $\boldsymbol{F}'$ denote the state transition matrix for CAV and HV, and $\boldsymbol{F}''$ denote the state transition matrix for passenger cars and trucks. The vehicle state transition probability matrix $\boldsymbol{F}$ for car–truck mixed heterogeneous traffic is then obtained as the Kronecker product of $\boldsymbol{F}'$ and $\boldsymbol{F}''$, expressed as:

$$\boldsymbol{F}' = \begin{bmatrix} \lambda_{\text{AA}} & \lambda_{\text{AH}} \\ \lambda_{\text{HA}} & \lambda_{\text{HH}} \end{bmatrix} \tag{1}$$

$$\boldsymbol{F}'' = \begin{bmatrix} 1-p_{\text{T}} & p_{\text{T}} \\ 1-p_{\text{T}} & p_{\text{T}} \end{bmatrix} \tag{2}$$

$$\boldsymbol{F} = \begin{bmatrix} \lambda_{\text{CAC–CAC}} & \lambda_{\text{CAC–CAT}} & \lambda_{\text{CAC–HC}} & \lambda_{\text{CAC–HT}} \\ \lambda_{\text{CAT–CAC}} & \lambda_{\text{CAT–CAT}} & \lambda_{\text{CAT–HC}} & \lambda_{\text{CAT–HT}} \\ \lambda_{\text{HC–CAC}} & \lambda_{\text{HC–CAT}} & \lambda_{\text{HC–HC}} & \lambda_{\text{HC–HT}} \\ \lambda_{\text{HT–CAC}} & \lambda_{\text{HT–CAT}} & \lambda_{\text{HT–HC}} & \lambda_{\text{HT–HT}} \end{bmatrix} = \boldsymbol{F}' \otimes \boldsymbol{F}'' = \begin{bmatrix} \lambda_{\text{AA}}(1-p_{\text{T}}) & \lambda_{\text{AA}}p_{\text{T}} & \lambda_{\text{AH}}(1-p_{\text{T}}) & \lambda_{\text{AH}}p_{\text{T}} \\ \lambda_{\text{AA}}(1-p_{\text{T}}) & \lambda_{\text{AA}}p_{\text{T}} & \lambda_{\text{AH}}(1-p_{\text{T}}) & \lambda_{\text{AH}}p_{\text{T}} \\ \lambda_{\text{HA}}(1-p_{\text{T}}) & \lambda_{\text{HA}}p_{\text{T}} & \lambda_{\text{HH}}(1-p_{\text{T}}) & \lambda_{\text{HH}}p_{\text{T}} \\ \lambda_{\text{HA}}(1-p_{\text{T}}) & \lambda_{\text{HA}}p_{\text{T}} & \lambda_{\text{HH}}(1-p_{\text{T}}) & \lambda_{\text{HH}}p_{\text{T}} \end{bmatrix} \tag{3}$$

The state transition probabilities in $\boldsymbol{F}'$ are defined as follows [37]:

$$\lambda_{\text{AH}} = \begin{cases} (1-p_{\text{A}})(1-O), & O \ge 0 \\ (1-p_{\text{A}}) + O\{(1-p_{\text{A}}) - \min[1,(1-p_{\text{A}})/p_{\text{A}}]\}, & O < 0 \end{cases} \tag{4}$$

$$\lambda_{\text{HA}} = \begin{cases} p_{\text{A}}(1-O), & O \ge 0 \\ p_{\text{A}} + O\{p_{\text{A}} - \min[1,p_{\text{A}}/(1-p_{\text{A}})]\}, & O < 0 \end{cases} \tag{5}$$

$$\lambda_{\text{AA}} = 1 - \lambda_{\text{AH}} \tag{6}$$

$$\lambda_{\text{HH}} = 1 - \lambda_{\text{HA}} \tag{7}$$

where $\lambda_{sr}$ is the transition probability of a vehicle of type $r$ following a vehicle of type $s$. In heterogeneous traffic flow, $s$ and $r \in${A, H}, A represents a CAV, and H represents an HV. In car-truck mixed heterogeneous traffic flow, $s$ and $r \in${CAC, CAT, HC, HT}.

When the $n$-th ($n > 1$) vehicle in the traffic flow is an HC, the probabilities of it following HC, HT, CAC, and CAT are determined based on the state space and the corresponding transition probability matrix. These probabilities are denoted as $P_{\text{HC-HC}}$, $P_{\text{CAC-HC}}$, $P_{\text{HT-HC}}$, and $P_{\text{CAT-HC}}$, respectively.

$$\begin{cases} P_{\text{HC–HC}} = P(S_{n-1} = \text{HC}, S_n = \text{HC}) = P(S_{n-1} = \text{HC})P(S_n = \text{HC}|S_{n-1} = \text{HC}) = p_{\text{HC}}\lambda_{\text{HC–HC}} \\ P_{\text{CAC–HC}} = P(S_{n-1} = \text{CAC}, S_n = \text{HC}) = P(S_{n-1} = \text{CAC})P(S_n = \text{HC}|S_{n-1} = \text{CAC}) = p_{\text{CAC}}\lambda_{\text{CAC–HC}} \\ P_{\text{HT–HC}} = P(S_{n-1} = \text{HT}, S_n = \text{HC}) = P(S_{n-1} = \text{HT})P(S_n = \text{HC}|S_{n-1} = \text{HT}) = p_{\text{HT}}\lambda_{\text{HT–HC}} \\ P_{\text{CAT–HC}} = P(S_{n-1} = \text{CAT}, S_n = \text{HC}) = P(S_{n-1} = \text{CAT})P(S_n = \text{HC}|S_{n-1} = \text{CAT}) = p_{\text{CAT}}\lambda_{\text{CAT–HC}} \end{cases} \tag{8}$$

where $S_n$ is the vehicle type of the $n$-th vehicle in the traffic flow.

Similarly, the car-following probabilities for HT are determined and denoted as $P_{\text{HC-HT}}$, $P_{\text{CAC-HT}}$, $P_{\text{HT-HT}}$, and $P_{\text{CAT-HT}}$.

When the $n$-th ($n > 1$) vehicle in the traffic flow is a CAC, six possible CF types are considered. Let $P_{\text{PL1}}$ denote the probability that a CAV is the leading vehicle of a CAV platoon, expressed as:

$$\begin{aligned} P_{\text{PL1}} &= P(S_{n-1} = \text{HC}, S_n = \text{CAC}) + P(S_{n-1} = \text{HT}, S_n = \text{CAC}) + P(S_{n-1} = \text{HC}, S_n = \text{CAT}) + \\ &\quad P(S_{n-1} = \text{HT}, S_n = \text{CAT}) + P(S_{n-R+1} = \text{PL1}, S_{n-R+2} = \text{CAV}, \cdots, S_n = \text{CAV}) + \\ &= p_{\text{HC}}\lambda_{\text{HC–CAC}} + p_{\text{HT}}\lambda_{\text{HT–CAC}} + p_{\text{HC}}\lambda_{\text{HC–CAT}} + p_{\text{HT}}\lambda_{\text{HT–CAT}} + P_{\text{PL1}}(\lambda_{\text{AA}})^R \end{aligned} \tag{9}$$

Based on Eq. (9), the expression for $P_{\text{PL1}}$ can be derived as:

$$P_{\text{PL1}} = \begin{cases} 1/R, & p_{\text{A}} = 1 \text{or} O = 1 \\ (p_{\text{HC}}\lambda_{\text{HC–CAC}} + p_{\text{HT}}\lambda_{\text{HT–CAC}} + p_{\text{HC}}\lambda_{\text{HC–CAT}} + p_{\text{HT}}\lambda_{\text{HT–CAT}})/\left[1-(\lambda_{\text{AA}})^R\right], & \text{other} \end{cases} \tag{10}$$

Accordingly, the probabilities of CAC following HC, HT, the tail vehicles CAC or CAT of a CAV platoon, as well as CAC following CAC or CAT within a CAV platoon, can be defined as $P_{\text{HC-CAC}}$, $P_{\text{HT-CAC}}$, $P_{\text{PL_C-CAC}}$, $P_{\text{PL_T-CAC}}$, $P_{\text{PL_C-C}}$, and $P_{\text{PL_T-C}}$.

$$
\begin{cases}
P_{\mathrm{HC-CAC}} = P(S_{n-1} = \mathrm{HC}, S_n = \mathrm{CAC}) = p_{\mathrm{HC}}\lambda_{\mathrm{HC-CAC}} \\
P_{\mathrm{HT-CAC}} = P(S_{n-1} = \mathrm{HT}, S_n = \mathrm{CAC}) = p_{\mathrm{HT}}\lambda_{\mathrm{HT-CAC}} \\
P_{\mathrm{PL_C-CAC}} = P(S_{n-R+1} = \mathrm{PL1}, S_{n-R+2} = \mathrm{CAV}, \cdots, S_n = \mathrm{CAV})(1-p_{\mathrm{T}})^2 = P_{\mathrm{PL1}}(\lambda_{\mathrm{AA}})^R(1-p_{\mathrm{T}})^2 \\
P_{\mathrm{PL_T-CAC}} = P(S_{n-R+1} = \mathrm{PL1}, S_{n-R+2} = \mathrm{CAV}, \cdots, S_n = \mathrm{CAV})p_{\mathrm{T}}(1-p_{\mathrm{T}}) = P_{\mathrm{PL1}}(\lambda_{\mathrm{AA}})^R p_{\mathrm{T}}(1-p_{\mathrm{T}}) \\
P_{\mathrm{PL_C-C}} = \sum_{m=2}^{R} P(S_{n-m+1} = \mathrm{PL1}, S_{n-m+2} = \mathrm{CAV}, \cdots, S_n = \mathrm{CAV})(1-p_{\mathrm{T}})^2 = \sum_{m=2}^{R} P_{\mathrm{PL1}}(\lambda_{\mathrm{AA}})^{m-1}(1-p_{\mathrm{T}})^2 \\
P_{\mathrm{PL_T-C}} = \sum_{m=2}^{R} P(S_{n-m+1} = \mathrm{PL1}, S_{n-m+2} = \mathrm{CAV}, \cdots, S_n = \mathrm{CAV})p_{\mathrm{T}}(1-p_{\mathrm{T}}) = \sum_{m=2}^{R} P_{\mathrm{PL1}}(\lambda_{\mathrm{AA}})^{m-1}p_{\mathrm{T}}(1-p_{\mathrm{T}})
\end{cases}
\tag{11}
$$

Similarly, the car-following probabilities for CAT are determined and denoted as $P_{\mathrm{HC\text{-}CAT}}$, $P_{\mathrm{HT\text{-}CAT}}$, $P_{\mathrm{PL_C\text{-}CAT}}$, $P_{\mathrm{PL_T\text{-}CAT}}$, $P_{\mathrm{PL_C\text{-}T}}$, and $P_{\mathrm{PL_T\text{-}T}}$.

By the definition of traffic density, the density of car-truck mixed heterogeneous traffic flow is defined as the reciprocal of the average headway, as follows:

$$
k = \frac{1}{\sum_{n=1}^{N} P_n H_n} \tag{12}
$$

where $k$ is the density of car-truck mixed heterogeneous traffic flow, $P_n$, and $H_n$ are the probability and equilibrium headway of the $n$-th CF type, $N$ is the total number of CF types.

Building upon this foundation, the Newell model [38] is adopted to characterize the CF behavior of HV. the ACC model is utilized to describe the CF behavior of a CAV following an HV, and the CACC model is utilized to describe the CF behavior of a CAV following another CAV. The relationships between headway ($H$) and speed ($v$) for each CF model are established, as shown in Table 1.

Based on the relationship between headway and speed, the functional relationship between speed and density of car–truck mixed heterogeneous traffic flow is established.

$$
v = \frac{1}{k\sum_{n=1}^{N} P_n T_n} - \frac{\sum_{n=1}^{N} P_n d_n}{\sum_{n=1}^{N} P_n T_n} \tag{13}
$$

The relationship between flow and density of car-truck mixed heterogeneous traffic flow is established.

$$
q = \frac{1}{\sum_{n=1}^{N} P_n T_n} - \frac{\sum_{n=1}^{N} P_n d_n}{\sum_{n=1}^{N} P_n T_n} k \tag{14}
$$

Eq. (14) is employed to characterize the congested regime of the fundamental diagram for car-truck mixed heterogeneous traffic flow, and is supplemented by the free-flow regime to yield the complete fundamental diagram, as shown in Fig. 3. In the figure, $v_{\mathrm{f}}$ represents the free-flow speed, $q_{\max}$ represents the maximum capacity, $k_{\mathrm{c}}$ represents the optimal density at maximum capacity, $k_{\mathrm{J}}$ represents the jam density, $w_{\mathrm{c}}$ represents the backward wave speed under congested conditions. Based on Eqs. (12) - (14), these parameters are further derived.

$$
v_{\mathrm{f}} = \sum_{n=1}^{N} P_n v_{\mathrm{f}}^n \tag{15}
$$

$$
k_{\mathrm{c}} = \frac{1}{\sum_{n=1}^{N} P_n \left(v_{\mathrm{f}}^n T_n + d_n\right)} \tag{16}
$$

$$
q_{\max} = \sum_{n=1}^{N} \frac{v_{\mathrm{f}}^n}{P_n \left(v_{\mathrm{f}}^n T_n + d_n\right)} \tag{17}
$$

$$
k_{\mathrm{J}} = \frac{1}{\sum_{n=1}^{N} P_n d_n} \tag{18}
$$

**Table 1**
Calculation formulas for the equilibrium headways in CF models.

| CF type | CF model | Headway | Parameters interpretation |
|---|---|---|---|
| HV following HV or CAV | Newell model | $H = vT_{\mathrm{H}} + d_{\mathrm{H}}$ | $T_{\mathrm{H}}$, $d_{\mathrm{H}}$ are the time headway and minimum safety headway of Newell model |
| CAV following HV | ACC model | $H = vT_{\mathrm{A}} + d_{\mathrm{A}}$ | $T_{\mathrm{A}}$, $d_{\mathrm{A}}$ are the time headway and minimum safety headway of ACC model |
| CAV following CAV | CACC model | $H = vT_{\mathrm{C}} + d_{\mathrm{C}}$ | $T_{\mathrm{C}}$, $d_{\mathrm{C}}$ are the time headway and minimum safety headway of CACC model |

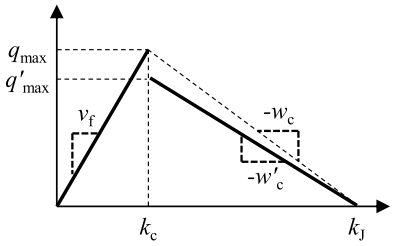



**Fig. 3.** The fundamental diagram of the CTM model.

$$w_c = -\frac{\sum_{n=1}^{N} P_n d_n}{\sum_{n=1}^{N} P_n T_n} \tag{19}$$

In addition, frequent fluctuations in vehicle speed and lane-changing behavior induce traffic disturbances in bottleneck areas, leading to dual capacity phenomena. Accordingly, building upon existing studies [39], an optimized fundamental diagram model of traffic flow incorporating dual capacity phenomena is proposed as follows: (1) When traffic density remains below the critical threshold $k_c$, traffic flow increases smoothly, and the maximum capacity remains stable; (2) When the traffic density exceeds the critical threshold $k_c$, capacity undergoes a sharp decline. The variations in capacity and backward wave speed before and after the sudden decline are as follows.

$$q'_{max} = (1-\alpha) q_{max} \tag{20}$$

$$w'_c = (1-\alpha) w_c \tag{21}$$

where $q'_{max}$ is the capacity after attenuation, $w'_c$ is the reverse wave velocity after attenuation, and $\alpha$ is the attenuation coefficient, with a value range of 5–20 %.

#### 4.1.2. The CTM of merging area

The CTM is a macroscopic traffic flow model based on space-time discretization. It divides the roadway into uniform cells of length $\Delta L$ and simulates the dynamic evolution of traffic flow, density, and other variables using a fixed time step $\Delta t$. Due to its high computational efficiency, simplicity, and ease of secondary development, CTM is well-suited for traffic simulation and prediction. Therefore, this study adopts the CTM to simulate car–truck mixed heterogeneous traffic flow in freeway merging areas. Each lane within the research section is partitioned into $N_C$ cells. Following the driving direction, lane cells are indexed from upstream to downstream by $i \in \{1, 2, \ldots, N_C\}$. Mainline lanes are denoted by $j \in \{1, 2\}$, numbered from the outer to the inner lane, and ramp lanes are similarly indexed by $i'$. The structure of road cells and the corresponding traffic flow directions are illustrated in Fig. 4.

As shown in Fig. 4, the mainline cells are classified into basic segment cells and merging point cells based on the traffic flow direction of each cell. According to the principle of traffic flow conservation, the number of vehicles in a cell at the next time step is determined by the current number of vehicles, the inflow from the upstream through lanes and adjacent lane changes, and the outflow to the downstream through lanes and adjacent lane changes, as follows:

$$n_{i,j}(t+1) = n_{i,j}(t) + \sum_{\forall k \in \{j, j'\}} y^{j}_{i-1,k}(t) - \sum_{\forall k \in \{j, j'\}} y^{k}_{i,j}(t) \tag{22}$$

where $n_{i,j}(t)$ is the number of vehicles in cell $(i, j)$ at time step $t$, $y^{j}_{i-1,k}(t)$ is the number of vehicles flowing from cell $(i\text{-}1, k)$ into cell $(i, j)$, $k \in \{j, j'\}$, $j'$ is an adjacent lane.

In the CTM, the traffic flow between adjacent cells is jointly determined by the sending capacity of the upstream cell and the receiving capacity of the downstream cell. The sending capacity of the upstream cell $(i\text{-}1, k)$, denoted as $S_{i-1,k}(t)$, depends on the number of vehicles currently in the cell and the maximum capacity per time step $Q_{max}$, and is expressed as:

$$S_{i-1,k}(t) = \min\left[n_{i-1,k}(t), Q_{max}\right] \tag{23}$$

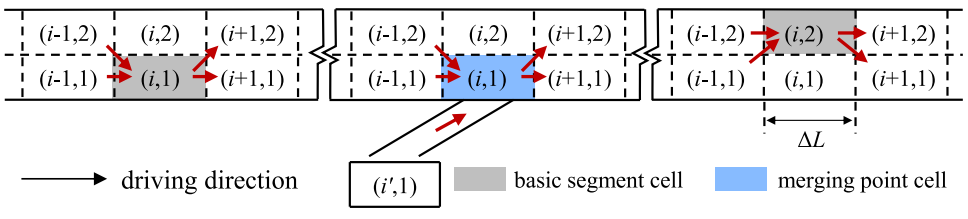



**Fig. 4.** Lane-revel cell transmission model.

The receiving capacity of the downstream cell ($i$, $k$), denoted as $R_{i,k}(t)$, depends on the maximum capacity per time step and the available space in the cell, and is expressed as:

$$R_{i,k}(t) = \min\left\{Q_{\max}, \frac{w_c}{v_f}\left[k_J \Delta L - n_{i,k}(t)\right]\right\} \tag{24}$$

Building on this, let the sending capacity from cell ($i$-1, $k$) to cell ($i$, $j$) be denoted as $S^j_{i-1,k}(t)$, and the corresponding expected proportion be denoted as $p^j_{i-1,k}$. The $p^j_{i-1,k}$ is determined by the current traffic states of the cell and its neighboring cells and is calculated as follows:

$$S^j_{i-1,k}(t) = p^j_{i-1,k}(t) S_{i-1,j}(t) \tag{25}$$

$$\begin{cases} p^j_{i-1,k} = \dfrac{\max\left[0, v_{i-1,j}(t) - v_{i-1,k}(t)\right]}{\tau_{LC} v_f}, & k \neq j \\ p^j_{i-1,k} = 1 - \sum\limits_{\forall k' \neq k} p^{k'}_{i-1,k}, & k = j \end{cases} \tag{26}$$

where $v_{i-1,j}(t)$ is the speed of the vehicle in the cell ($i$-1, $j$), $\tau_{LC}$ is the time coefficient.

According to Eqs. (23) - (26), the actual number of vehicles flowing into and out of cell ($i$,$j$) can be determined as follows:

$$\gamma^j_{i-1,k} = \min\left[1, \frac{R_{i,j}(t)}{\sum\limits_{\forall k \in \{j,j'\}} S^j_{i-1,k}(t)}\right] \tag{27}$$

$$y^j_{i-1,k}(t) = \gamma^j_{i-1,k} S^j_{i-1,k}(t) \tag{28}$$

where $\gamma^j_{i-1,k}$ is the proportion of vehicles that flow from cell ($i$-1, $k$) to cell ($i$, $j$), $k \in \{j, j'\}$.

Then, the cell density is determined as $k_{i,j}(t+1) = n_{i,j}(t+1)/\Delta L$, and the cell speed is subsequently calculated as follows:

$$v_{i,j}(t+1) = \begin{cases} v_f, & k_{i,j}(t+1) \leq k_c \\ \dfrac{\left[k_J - k_{i,j}(t+1)\right] w_c}{k_{i,j}(t+1)}, & k_{i,j}(t+1) > k_c \end{cases} \tag{29}$$

Similarly, $y^k_{i,j}(t)$ is calculated. The number of vehicles and speed for each cell, including both basic segment cells and merging point cells, is updated according to Eq. (22).

### 4.2. The VSL control of freeway merging area

#### 4.2.1. The VSL optimization function

The VSL control method is an ATM strategy to alleviate congestion at freeway bottlenecks. The primary objective of the VSL control method is to ensure traffic safety and improve operational efficiency, thereby mitigating congestion. The fundamental diagram of CTM model for VSL control is illustrated in Fig. 5. Under VSL control, if the speed limit exceeds the free-flow speed of a given road segment, the fundamental diagram of the CTM model remains unchanged. However, when the speed limit is set below the free-flow speed, the maximum speed is reduced to the specified limit, reducing the capacity of the controlled area. In this context, the speed limit is denoted as $v_{sl}$, the capacity as $q_{sl}$, and the corresponding critical density as $k_{sl}$.

The formulas of $k_{sl}$ and $q_{sl}$ under VSL control are as follows:

$$k_{sl} = w_c k_J / (w_c + v_{sl}) \tag{30}$$

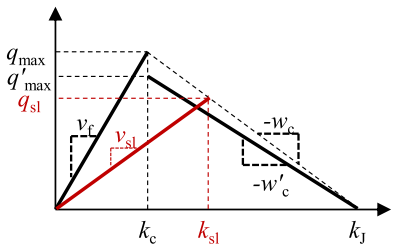



**Fig. 5.** The fundamental diagram of the CTM model for VSL control.

$$q_{sl} = k_{sl} v_{sl} \tag{31}$$

The heterogeneous operation of different vehicle types on the mainline and ramp within the merging area reduces traffic efficiency and poses significant safety risks. Consequently, this study comprehensively integrates efficiency and safety objectives, developing a dual-objective optimization model for VSL control in freeway merging areas.

*4.2.1.1. Efficiency objective.* Common performance indicators for traffic operational efficiency encompass total travel time (TTT), total turnover, and total travel delay. In this study, TTT is designated as the efficiency objective, with its formulation within the CTM model specified as follows:

$$TTT = \sum_{t=1}^{T_N} \sum_{i=1}^{N_C} \sum_{j=1}^{L_N} \left[ k_{i,j}(t) \Delta L \Delta t \right] \tag{32}$$

where $T_N$ is the number of time steps in the VSL control period, $N_C$ is the total number of cells, $k_{i,j}(t)$ is the density of cell $(i, j)$ in the $t$-th step. $L_N$ is the number of lanes.

*4.2.1.2. Safety objective.* Existing research on VSL control methods predominantly emphasizes efficiency objectives. In contrast, relatively few studies address safety objectives, primarily due to the lack of precise, quantitative safety evaluation indicators in macroscopic simulation models. Abdel-Aty et al. [40] collected data on density, speed, and flow, along with corresponding crash records, using loop detectors on a 0.5-mile segment of the I-4 freeway in Florida. Based on these data, they developed a statistical model using exponential regression to establish the relationship between crash risk and traffic flow parameters. Building on this approach, we consider the characteristics of car-truck mixed heterogeneous traffic flow in freeway merging areas, a simulation environment was constructed using the SUMO software to replicate such traffic scenarios. The following traffic indicators were collected: CAV penetration rate ($p_A$), truck mixing rate ($p_T$), density ($k$), mainline speed ($v_M$), mainline flow ($q_M$), ramp flow ($q_R$), ramp speed ($v_R$), speed difference ($\Delta v$), time speed difference of mainline ($SVT$), and space speed difference of mainline ($SVD$). In addition, collision risks were assessed using traffic conflict ($TC$) indicators, and the number of conflicts under various traffic conditions was recorded. Subsequently, key operational indicators were identified through correlation analysis and multicollinearity testing. These

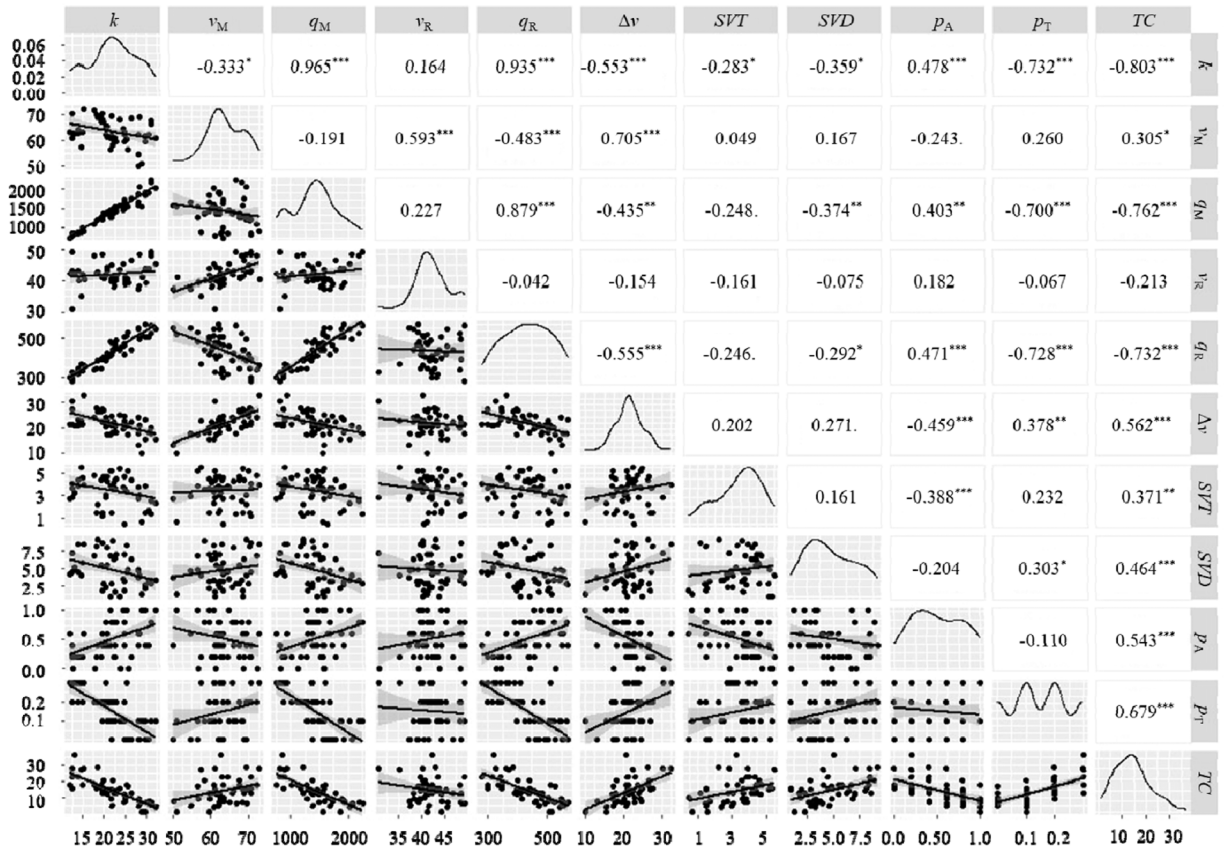



**Fig. 6.** Correlation analysis diagram of each variable.

indicators were used as independent variables in a regression model, with the number of traffic conflicts as the dependent variable, to construct a collision risk prediction model. This model was then adopted as the safety objective for the VSL control method.

The correlation analysis results, shown in Fig. 6, indicate strong associations between traffic conflicts and several traffic indicators, particularly $k$, $q_M$, $q_R$, $\Delta v$, $p_A$, and $p_T$. Some indicators demonstrated exact mathematical relationships. For instance, $\Delta v$ is directly related to $v_M$ and $v_R$, leading to perfect multicollinearity. Consequently, $v_M$ and $v_R$ were excluded from further analysis. The remaining variables were tested for multicollinearity using the variance inflation factor (VIF). As shown in Table 2, VIF values for $k$, $q_M$, and $q_R$ all exceeded 10, suggesting a high degree of multicollinearity. Given that density is strongly correlated with flow, $q_M$ and $q_R$ were also excluded. Ultimately, six key feature indicators were retained: $k$, $\Delta v$, $SVT$, $SVD$, $p_A$, and $p_T$.

Currently, researchers commonly use count models to construct traffic risk prediction models [41]. Following this approach, poisson regression (PR), negative binomial regression (NBR), zero-inflated negative binomial regression (ZNBR), and bayesian negative binomial regression (BNBR) were applied for regression analysis. The model fit comparison results are shown in Table 3.

As shown in Table 3, the NBR model outperforms the PR and ZNBR models in terms of AIC, BIC, and RMSE, with values of 266.03, 297.34, and 3.67, respectively. This indicates that the NBR model provides the best overall fit. The main reason is that the dependent variable exhibits overdispersion, meaning its variance exceeds the mean. The NBR model addresses this issue by relaxing the PR model assumption of equal mean and variance. Although the traffic conflict data includes zero values, their proportion is relatively low, making ZNBR less effective than NBR. The BNBR builds on the NBR model by incorporating Bayesian parameter estimation. In this study, the Metropolis-Hastings algorithm of the MCMC method was used to sample the parameters of the BNBR model. Two types of prior distributions were tested: (1) assuming the NBR parameters follow a normal distribution and (2) assuming they follow a uniform distribution. In both cases, the logarithm of the overdispersion parameter was assumed to follow a normal distribution. According to Table 3, the BNBR model with normal priors achieved a better fit than the model with uniform priors. However, the RMSE values of both BNBR models were higher than that of the standard NBR model, indicating that NBR still provides the best overall fit. Therefore, the NBR model is selected as the collision risk prediction model in this study.

Based on the final fitting results, the collision risk prediction model is established as follows:

$$TC = \exp(2.458 - 0.035k + 0.017\Delta v + 0.076SVT + 0.043SVD - 0.348p_A + 1.329p_T) \tag{33}$$

The model's null deviance is 197.611, and the residual deviance is 47.578. The relatively low residual deviance indicates a good model fit. Owing to its strong explanatory power, the model is adopted as the basis for collision risk prediction in the merging area. The independent variables are subsequently instantiated via the CTM framework to compute cell-level collision risk forecasts over a finite prediction horizon within the merging area. The calculation is as follows:

$$TC = \exp\begin{pmatrix} 2.458 - 0.035k_{i,j}(t) + 0.017\left|v_{i,j}(t) - v_{i',1}(t)\right| + 0.076\left|v_{i,j}(t) - v_{i,j}(t-1)\right| \\ +0.043\left|v_{i,j+1}(t) - v_{i,j}(t)\right| - 0.348p_A + 1.329p_T \end{pmatrix} \tag{34}$$

where $k_{i,j}(t)$ and $v_{i,j}(t)$ are the density and speed of cell $(i, j)$ at time step $t$, respectively.

In summary, the optimization function of the VSL control method, which integrates both efficiency and safety considerations, is defined as follows:

$$\begin{cases} TTT = \sum_{t=1}^{T_N}\sum_{i=1}^{N_C}\sum_{j=1}^{L_N}\left[k_{i,j}(t)\Delta L\Delta t\right] \\ TCR = \sum_{t=1}^{T_N} TC(t) \end{cases} \tag{35}$$

where $TCR$ is total collision risk.

To ensure the practical feasibility of the VSL control method, it is necessary to appropriately constrain the efficiency and safety of optimization functions. Specifically: (1) To ensure the stability and safety of traffic flow, the speed limit must remain within a specified range and be a multiple of 5 km/h; (2) To enhance driver compliance, the difference between speed limit values of adjacent road segments within the same control periods must not exceed a predefined threshold; (3) The change in the speed limit of the same road segment between adjacent control periods must not exceed the predefined threshold.

$$s.t.\begin{cases} v_{sl}^{\min} \le v_{sl}^{i,j}(t') \le v_{sl}^{\max} \\ \mathrm{mod}\left[v_{sl}^{i,j}(t'), 5\right] = 0 \\ \left|v_{sl}^{i,j}(t') - v_{sl}^{i-1,j}(t')\right| \le \Delta v_{\max} \\ \left|v_{sl}^{i,j}(t') - v_{sl}^{i,j}(t'-1)\right| \le \Delta v_{\max} \end{cases} \tag{36}$$

**Table 2**
VIF of each variable.

| variable | $k$ | $q_M$ | $q_R$ | $\Delta v$ | $SVT$ | $SVD$ | $p_A$ | $p_T$ |
|---|---|---|---|---|---|---|---|---|
| VIF | 38.108 | 20.579 | 9.199 | 1.920 | 1.261 | 1.234 | 2.001 | 2.905 |

**Table 3**
Comparison of goodness-of-fit of several models.

| Precision index | PR | NBR | ZNBR | BNBR | |
|---|---|---|---|---|---|
| | | | | normal distribution | uniform distribution |
| AIC | 287.97 | 266.03 | 281.08 | - | - |
| BIC | 303.26 | 297.34 | 309.76 | - | - |
| RMSE | 3.89 | 3.67 | 3.90 | 3.97 | 4.53 |
| DIC | - | - | - | 87.43 | 89.17 |

where $v_{sl}^{min}$ and $v_{sl}^{max}$ are the minimum and maximum speed limit values, respectively. $v_{sl}^{i,j}(t')$ is the speed limit for cell $(i, j)$ during the $t'$-th control period. $\Delta v_{max}$ is the maximum allowable difference in speed limits.

Eq. (35) and Eq. (36) together form a VSL control optimization function that simultaneously addresses both safety and efficiency objectives. The NSGA-II algorithm is a widely recognized and efficient method for multi-objective optimization. We employ the NSGA-II algorithm to solve the proposed optimization function, leveraging its efficient search capabilities to optimize the VSL control method in freeway merging areas.

#### 4.2.2. The MPC framework of VSL control

The MPC framework is a control architecture grounded in predictive modeling. Leveraging real-time traffic data, the CTM model is used to predict the future state of car-truck mixed heterogeneous traffic flows. This enables dynamic evaluation of upcoming traffic efficiency and safety, facilitating real-time adjustment of speed limits through closed-loop feedback control to enhance overall traffic efficiency and safety. Let $\Delta t$ denote the CTM time step, and $T_C$ represent the VSL control cycle, then each control cycle comprises $T_N = T_C/\Delta t$ steps. The detailed procedure is illustrated in Fig. 7 and can be described as follows:

(a) Real-time traffic state collection: Traffic data such as cell speed, density, and flow are continuously gathered via roadside units (RSUs) installed along the freeway.

(b) Traffic state prediction: Using current traffic data, the CTM model forecasts traffic conditions across the VSL control cycle.

(c) Objective optimization: Future traffic states are evaluated using total travel time (TTT) and traffic conflict risk (TCR) as performance metrics. A dual-objective VSL optimization function is formulated and solved using the NSGA-II algorithm to obtain the desired speed limit $v_{sl,e}^{i,j}(t'+1)$.

(d) Rolling optimization: The deviation between the observed density $k_{i,j}(t')$ and the predicted optimal density $k_{i,j}^{e}(t')$ is assessed. This deviation is then used to calculate a correction term $u_{i,j}(t'+1)$, which is applied to refine the speed limit, yielding a more accurate control value $v_{sl}^{i,j}(t'+1)$.

$$u_{i,j}(t'+1) = \eta\left[k_{i,j}(t') - k_{i,j}^{e}(t')\right] \tag{37}$$

where $\eta$ is the feedback coefficient for speed limit correction.

(e) Control implementation: The optimized speed limits are implemented via variable message signs and V2I communication, enabling closed-loop feedback control.

(f) State update: Traffic states are updated based on feedback, and the entire control process is iteratively repeated until the termination criterion is satisfied, thus ensuring periodic control.

## 5. Simulation experiment and discussion

### 5.1. Simulation experiment design

This study selects the interchange merging areas of the Suiman Freeway and the Harbin Ring Freeway in Heilongjiang Province,

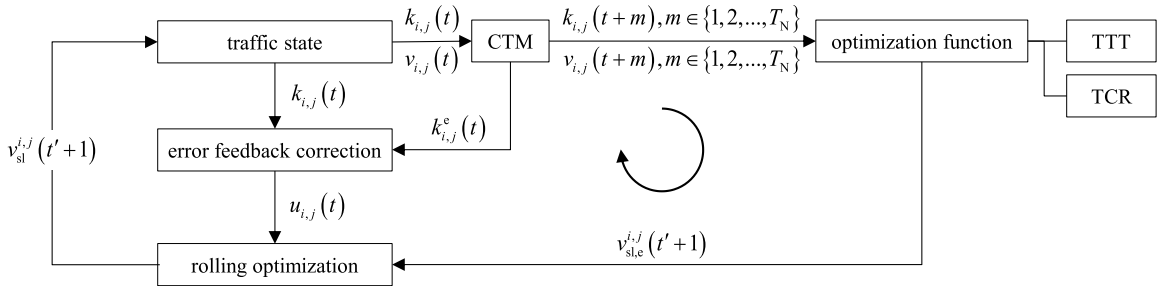



**Fig. 7.** MPC framework for VSL control.

China, as research scenarios to validate the proposed method and provide technical support for freeway traffic management authorities. The test section spans a total length of 4.8 km and comprises a one-way, two-lane road, with each cell measuring 400 m in length. The road segment is divided into 12 cells, with an entrance ramp positioned at Cell 9. A 1.2 km upstream section of the ramp is designated as the VSL area, and the research scenario is depicted in Fig. 8.

The total simulation duration is set to 120 min, with a simulation time step of 12 s, resulting in 600 simulation steps. The VSL control cycle is 5 min, equivalent to 25 simulation steps. The minimum and maximum speed limits are set at 50 km/h and 100 km/h, respectively, with a maximum allowable variation of 10 km/h between adjacent speed limits. Based on the measured traffic flow data during off-peak and peak periods in the freeway merging area, the input traffic flow for the simulation scenario was determined, as shown in Fig. 9.

The minimum safety headway comprises the minimum stopping distance and the vehicle length, with the lengths of a passenger car and a truck assumed to be 6.0 m and 12.0 m, respectively. According to the literature [42], the corresponding time headways and minimum stopping distances are presented in Table 4. In addition, the free-flow speeds are assumed to be 108 km/h for passenger cars and 70 km/h for trucks.

Based on the literature [35, 43–44], the parameters of the ACC and CACC models are defined as follows. For the ACC model, the expected headway time is 1.1 s for CAC and 1.6 s for CAT. For the CACC model, the expected headway times are reduced to 0.6 s for CAC and 0.9 s for CAT. When a CAV follows a CAV platoon, the CACC model remains applicable, with an added safety delay time of 0.26 s, resulting in modified expected headway times of 0.86 s for CAC and 1.16 s for CAT.

After calibrating the parameters of the car-following model, the fundamental diagram of car-truck mixed heterogeneous traffic flow is further analyzed. Fig. 10 illustrates the impact of the CAV maximum platoon size on capacity under varying CAV penetration rates and truck mixing rates. Fig. 10 show that capacity increases with the CAV maximum platoon size but gradually stabilizes. When the platoon size exceeds 7, the marginal improvement in capacity becomes negligible. Therefore, this study adopts 7 as the upper limit for the CAV maximum platoon size.

Fig. 11 illustrates the impact of the vehicle aggregation factor on capacity under varying CAV penetration rates and truck mixing rates. Fig. 11 show that capacity increases progressively with the rise of the aggregation factor. This trend is primarily due to stronger clustering of similar vehicle types, which enhances the likelihood of CAV platooning and, consequently, improves overall capacity. To analyze the baseline condition of traffic flow, the aggregation factor is set to 0 in the subsequent analysis, serving as a reference for exploring the characteristics of the fundamental diagram.

Fig. 12 illustrates the fundamental diagram of the car-truck mixed heterogeneous traffic flow under varying CAV penetration rates and truck mixing rates. Fig. 12 show that traffic flow initially increases with density before eventually decreasing. The maximum capacity and critical density increase as the CAV penetration rate rises, while they decline with a higher truck mixing rate. When the truck mixing rate is 0.2, the capacity for HV homogeneous flow is 2165 veh/h, whereas, for CAV homogeneous flow, it reaches 3435 veh/h, representing a 58.7 % increase. Similarly, when the truck mixing rate increases to 0.4, the capacity for CAV homogeneous flow drops to 3038 veh/h, reflecting an 11.6 % decrease compared to the capacity at a truck mixing rate of 0.2. These results suggest that increasing the CAV penetration rate positively impacts capacity and improves traffic conditions, whereas a higher truck mixing rate negatively affects capacity.

### 5.2. *Results analysis and discussion*

At the macroscopic aspect, TTT and TCR are primary evaluation indicators. At the microscopic aspect, average speed, average delay, and traffic conflict indicators (including time-exposed TTC (TET) and time-integrated TTC (TIT)) are employed to evaluate traffic efficiency and safety under the VSL strategy. Both TET and TIT are derived from the TTC, with a threshold of 3 s, as recommended in the relevant literature [45].

To evaluate the effectiveness of the proposed method, dual-objective optimization results under two strategies are compared, as presented in Table 5.

As shown in Table 5, the VSL strategy reduces both TTT and TCR compared to the baseline (no VSL) strategy, thereby enhancing traffic operational efficiency and promoting safety. Specifically, when the truck mixing rate is 0.2, and the CAV penetration rate reaches 0.5, the TTT decreases by 8.24 %, and the TCR decreases by 6.64 %. These results validate the effectiveness of the proposed method. Furthermore, as the CAV penetration rate increases, both TTT and TCR decrease, demonstrating the positive impact of CAVs on traffic flow and safety. In contrast, higher truck mixing rates increase TTT and TCR, indicating that heavy trucks negatively affect traffic efficiency and safety.

Regarding microscopic evaluation indicators, Fig. 13 and Fig. 14 compare average speed, average delay, and traffic conflict

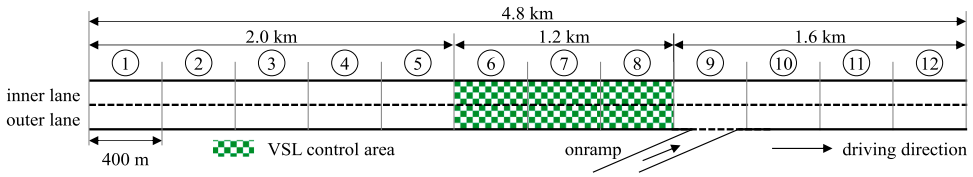



**Fig. 8.** Simulation experiment scenario.

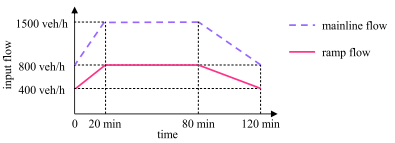
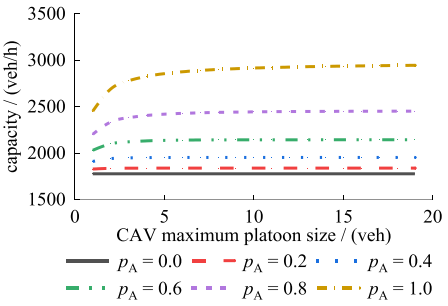


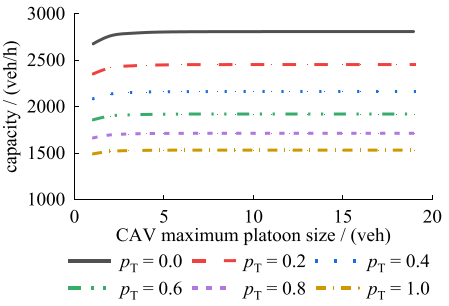

**Fig. 9.** Input flow of the mainline and the onramp.

**Table 4**
The time headway and minimum stopping distance in CF models.

| CF type | time headway (s) | minimum stopping distance (m) |
|---|---|---|
| HC following HC | 1.2 | 0.85 |
| HC following HT | 1.4 | 1.35 |
| HT following HC | 1.8 | 1.11 |
| HT following HT | 2.0 | 1.53 |

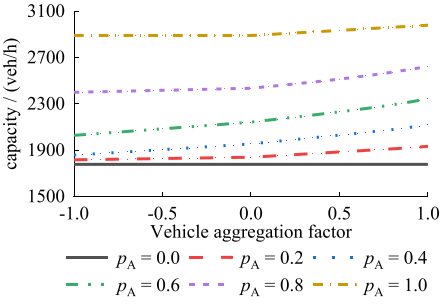



(1) The conditions for different CAV penetration rates ($p_T = 0.5$)

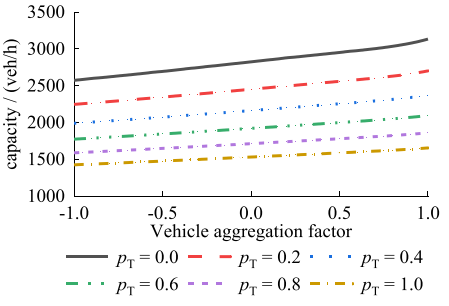



(2) The conditions for different truck mixing rates ($p_A = 0.5$)

**Fig. 10.** Impact of CAV maximum platoon size on capacity.




(1) The conditions for different CAV penetration rates ($p_T = 0.5$)




(2) The conditions for different truck mixing rates ($p_A = 0.5$)

**Fig. 11.** Impact of vehicle aggregation factor on capacity.

indicators TET and TIT.

Fig. 13 and Fig. 14 demonstrate that implementing the VSL strategy results in a higher average speed and lower average deceleration than the no VSL strategy. For example, when the truck mixing rate is 0.2, and the CAV penetration rate is 0.5, the VSL strategy increases the average speed by 5.48 % and reduces the average delay by 14.06 %. Furthermore, the TTT and TET indicators show improvement under the VSL strategy. Specifically, TET and TIT are reduced by 16.18 % and 30.29 %, respectively, compared to the no VSL strategy. These results confirm that the proposed VSL method effectively improves operational efficiency and safety.

A detailed analysis was conducted under the experimental conditions of a 0.2 truck mixing rate and a 0.5 CAV penetration rate. Fig. 15 presents the number of vehicles in each cell throughout the simulation. As traffic flow increases, the number of vehicles in the merging area and its upstream cells rises accordingly. Vehicle speeds drop sharply compared to no VSL strategy, resulting in severe congestion. By contrast, the application of the VSL strategy significantly reduces both the duration and spatial extent of congestion, further demonstrating its effectiveness in enhancing traffic efficiency.

The speed variation within the VSL area is illustrated in Fig. 16. During the congestion period, under no VSL strategy, upstream

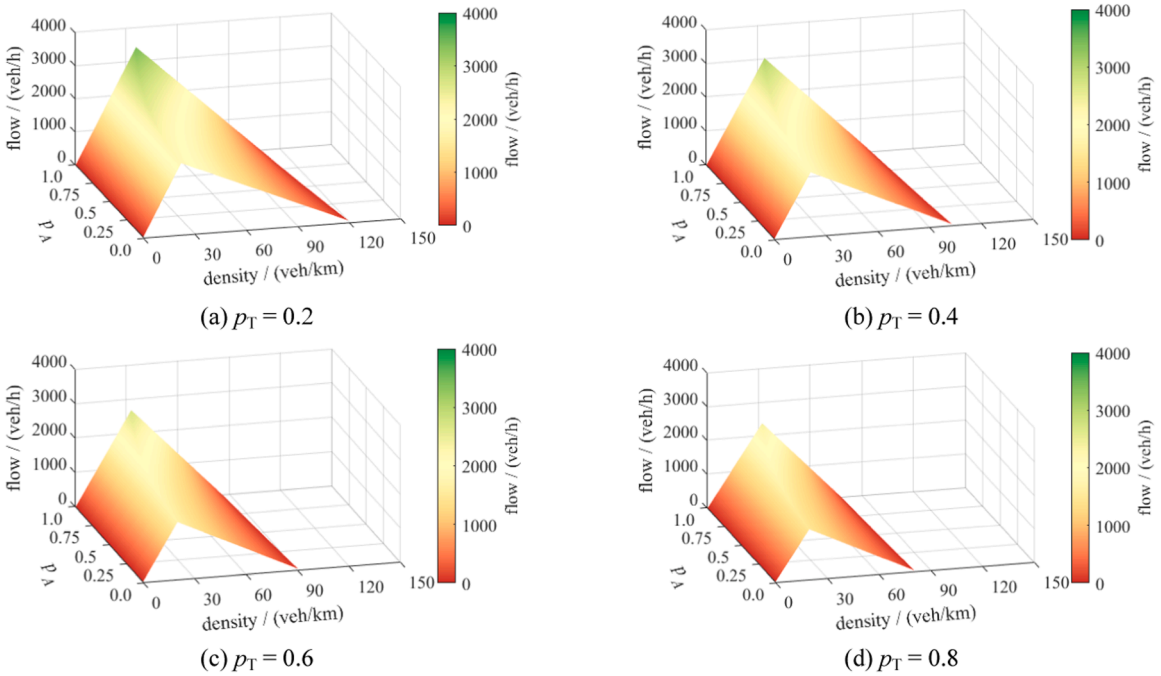



**Fig. 12.** The fundamental diagram of car-truck mixed heterogeneous traffic flow.

**Table 5**
Evaluation index results before and after VSL control.

| $p_T$ | $p_A$ | TTT / h | | | TCR / times | | |
|---|---|---|---|---|---|---|---|
| | | no VSL strategy | VSL strategy | reduction rate | no VSL strategy | VSL strategy | reduction rate |
| 0.2 | 0.25 | 223.11 | 198.93 | 10.84 % | 40.54 | 35.87 | 11.52 % |
| | 0.50 | 195.20 | 179.12 | 8.24 % | 31.64 | 29.54 | 6.64 % |
| | 0.75 | 155.24 | 154.44 | 0.52 % | 19.65 | 19.31 | 1.73 % |
| 0.4 | 0.25 | 301.28 | 255.15 | 15.31 % | 69.13 | 56.24 | 18.65 % |
| | 0.50 | 237.30 | 217.10 | 8.51 % | 48.45 | 43.32 | 10.59 % |
| | 0.75 | 189.55 | 184.12 | 2.86 % | 31.65 | 30.27 | 0.44 % |

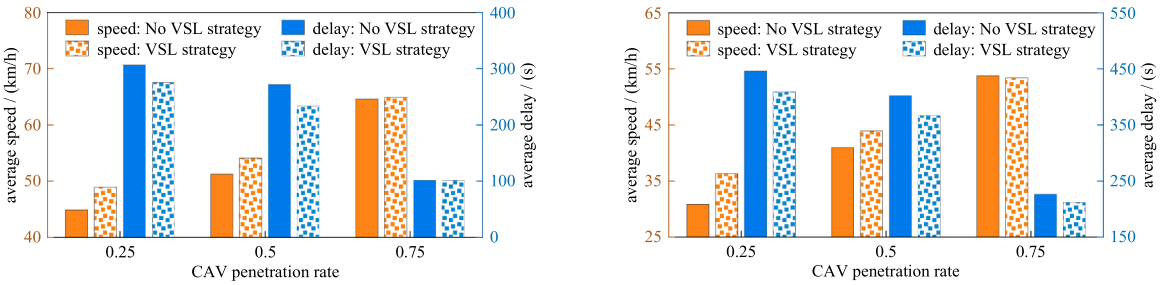
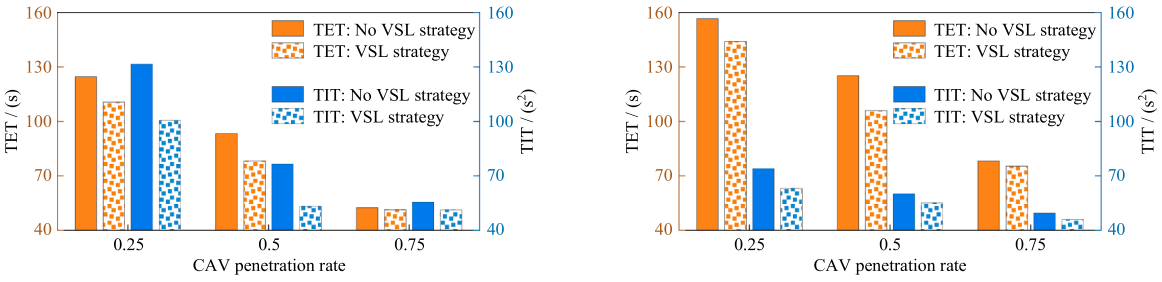



**Fig. 13.** Average speed and delay.

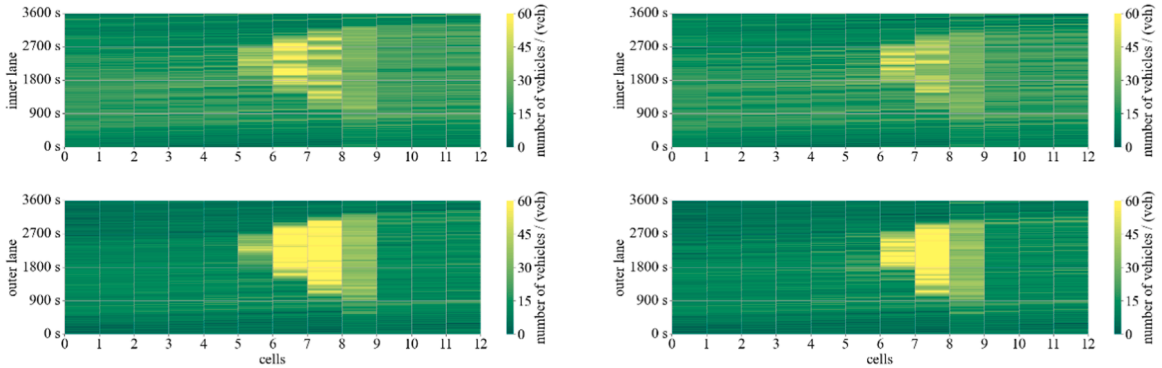
Cells 6, 7, and 8 in the merging area experience speeds below 25 km/h, with Cell 6 maintaining speeds under 25 km/h from 1100 to 3200 s, indicating severe congestion. In contrast, under VSL control, although Cells 7 and 8 also exhibit brief speeds below 25 km/h, the duration is significantly shorter, and the speed in Cell 6 remains consistently above 50 km/h.

The specific VSL control scheme is illustrated in Fig. 17. During periods of congestion, the speed limits in the VSL area are reduced, with the most significant decrease observed at Cell 8 near the merging area. This reduction is primarily due to intensified congestion in the merging area. Lower speed limits are implemented to regulate the upstream arrival rate of vehicles and achieve the dual objectives



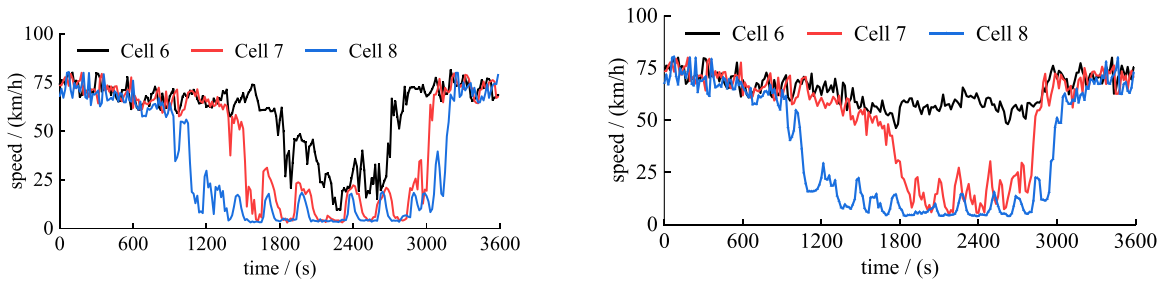

(a) Truck mixing rate is 0.2 (b) Truck mixing rate is 0.4

**Fig. 14.** TET and TIT indicators.




(a) no VSL strategy (b) VSL strategy

**Fig. 15.** Changes in the number of vehicles in each cell.




(a) no VSL strategy (b) VSL strategy

**Fig. 16.** Changes in the speed of VSL area cell.

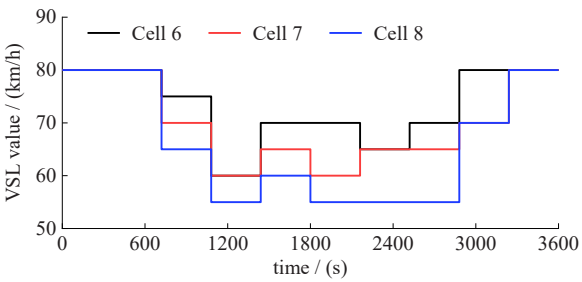



**Fig. 17.** VSL control scheme.

of minimizing travel delay and collision risk.

## 6. Conclusion

With the advancement of CAVs, heterogeneous traffic conditions are expected to emerge on freeways. Although previous studies have explored this topic, most existing macroscopic models fall short of capturing the dynamic evolution of car–truck mixed heterogeneous traffic. Moreover, prior research has primarily focused on the controllability of CAVs and improving traffic efficiency through ATM strategies, with limited consideration for traffic safety. No collision risk prediction model is designed explicitly for freeway merging areas under heterogeneous traffic conditions. As a result, a lack of VSL control methods that simultaneously address traffic efficiency and safety in freeway merging areas remains.

In this study, we develop a lane-level CTM for freeway merging areas under car–truck mixed heterogeneous traffic conditions to predict future traffic states over a finite time horizon. We collect data and construct a collision risk prediction model for merging areas based on a simulation environment replicating such traffic conditions. Building this, we formulate an MPC framework that integrates traffic state forecasting and speed limit optimization. As a result, a VSL control method tailored for freeway merging areas is proposed. The key findings of this study are summarized as follows:

a) This study comprehensively incorporates key factors such as the CAV penetration rate, truck mixing rate, aggregation factor, and the platooning characteristics of CACs and CATs to construct a car–truck mixed heterogeneous traffic flow CTM model. The results show that capacity increases with the maximum CAV platoon size but eventually stabilizes. Additionally, freeway capacity increases progressively with higher CAV penetration rates, while it declines with higher truck mixing rates.
b) A strong correlation is identified between traffic conflicts in freeway merging areas and variables such as density, speed differences, SVT, SVD, CAV penetration rate, and truck mixing rate. A collision risk prediction model is developed using negative binomial regression based on these variables. The model demonstrates a high goodness-of-fit, indicating its effectiveness in predicting collision risk in merging areas.
c) A VSL control method for freeway merging areas is proposed within an MPC framework to enhance traffic efficiency and safety. Simulation results indicate that, compared to no VSL control condition, the proposed method reduces total travel time and collision risk at the macroscopic aspect. In contrast, at the microscopic aspect, it increases average speed, reduces delay, and improves traffic conflict indicators such as TET and TIT. These findings verify the effectiveness and feasibility of the proposed approach.

The proposed MPC-based VSL control method significantly enhances traffic flow efficiency and reduces collision risk. It provides a scientifically robust and practically applicable decision-making and management tool for freeway traffic management. Nevertheless, several aspects warrant further refinement. Future research should incorporate additional factors such as CAV platoon coordination characteristics, adverse weather conditions, and driver lane preference behaviors to improve model accuracy. Moreover, extending the approach to more complex scenarios, such as continuous ramp merging areas and weaving sections, will enhance its applicability and robustness in real-world traffic environments.

## CRediT authorship contribution statement

**Wenzhi Wang:** Writing – original draft, Data curation. **MENG fengwei:** Writing – original draft, Visualization, Data curation. **Guozhu Cheng:** Writing – review & editing, Methodology, Funding acquisition. **Yongsheng Chen:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

Data will be made available on request.

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